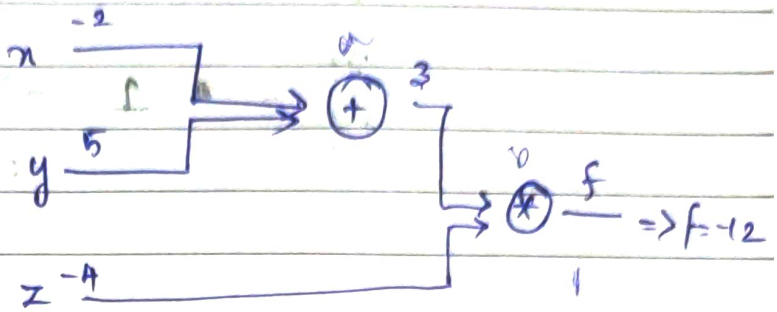
Backpropagation Algorithm:-

computational Graph

$f = (x+y)z$



$\frac{\partial f}{\partial x} = -4$

$\frac{\partial f}{\partial y} = -4$

$\frac{\partial f}{\partial z} = 3$

$\frac{\partial f}{\partial x} = xz + yz$ → x becomes 1 ⇒ constant becomes zero

$= z$ ⇒ -4

$\partial f / \partial y = (x+y)z$

$= xz + yz$ ⇒ -4

xz becomes constant and gone to zero; y becomes 1.

$\frac{\partial f}{\partial z} = (x+y)z$

$= xz + yz$ z becomes 1.

$= x + y$

$= 3$

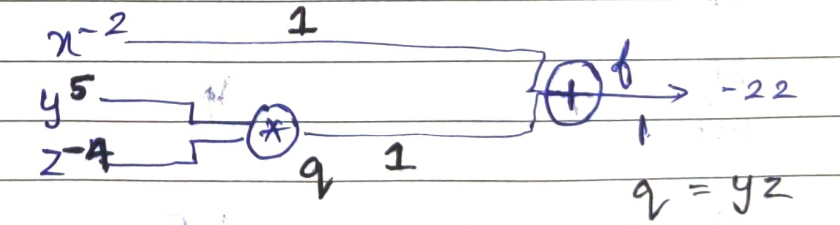
forward pass → ; ← Backward pass

$f = x + yz = -22$

$x = -2$

$y = 5$

$z = -4$



$q = yz$

$\frac{\partial f}{\partial x} = 1 + yz = 1 + (-20) = -19$ $\frac{\partial q}{\partial y} = z$

$\frac{\partial f}{\partial y} = z = -4$ $\frac{\partial q}{\partial z} = y$

$\partial f / \partial z = y = 5$

Local Gradient

$f = x + q$

$\frac{\partial f}{\partial x} = 1$ → local × global

$= \partial f/\partial x * 1 = 1 * 1$

$\frac{\partial f}{\partial q} = 1 \Rightarrow$ local × global

$= 1 \times 1$

$= 1$

$\frac{\partial f}{\partial y}$ = local × global.

$\frac{\partial q}{\partial y} *$ global

$= -4 \times 1$

Whatever coming into the gate in return is the Global

$f = (xy + w)v \quad = f \Rightarrow xyv + wv$

$x = 1 \quad y = 2 \quad w = 3 \quad v = 4.$

$f = xyv + wv$

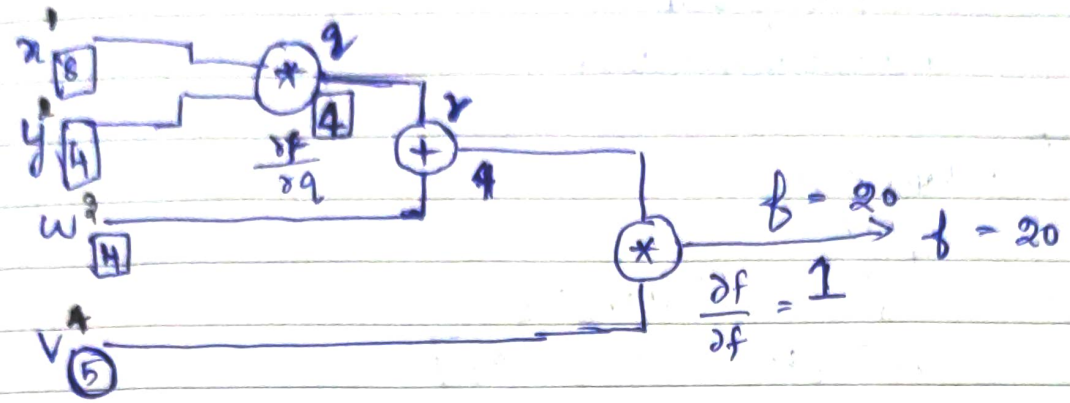
$\frac{\partial f}{\partial x} = yv = 8$

$\frac{\partial f}{\partial y} = xv = 4$

$\frac{\partial f}{\partial w} = v = 4$

$\frac{\partial f}{\partial v} = xy + w = 0.5$

Computational.
(Expression) Graph



$$\frac{\partial f}{\partial v} = xy + w$$

$$f = r * v$$

$$\frac{\partial f}{\partial r} = v$$

$$\frac{\partial f}{\partial v} = r \ (=) \ 5$$

$$q = x * y$$

$$\frac{\partial r}{\partial q} = 1$$

$$\frac{\partial f}{\partial x} = y \times 4 = 2 \times 4 = 8$$

$$\frac{\partial f}{\partial y} = x \times 4 = 1 \times 4 = 4$$

27/07/2021

Lecture => 7

→ Back propagation.

→ Activation function.

$$z_1^{[1]} = w_{11} x_1 + w_{12} x_2$$

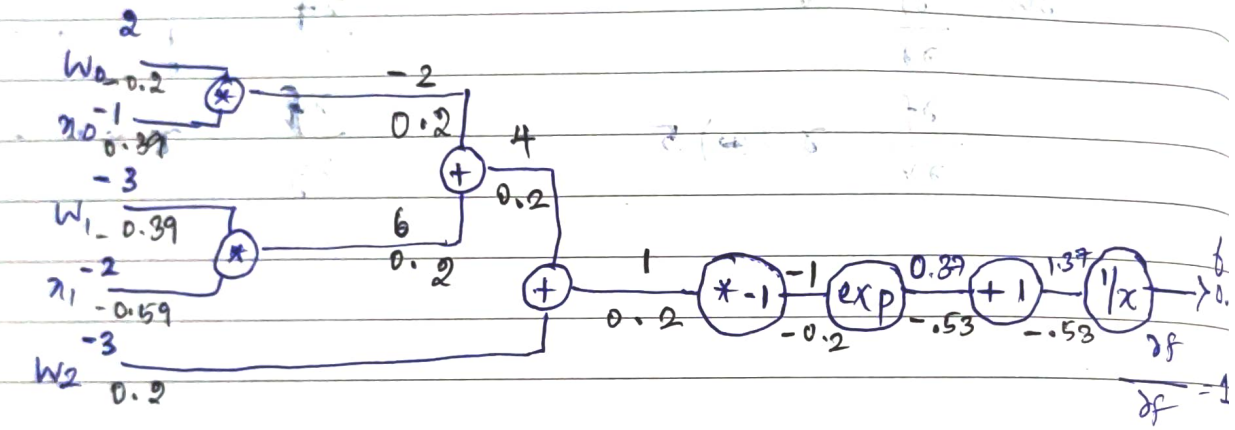
$$a_1^{[1]} = z_1^{[1]}$$

$$z_1^{[2]} = w_{11}^{[2]} z_1^{[1]} + w_{12} z_2^{[1]}$$

$$= w_{11}^{[2]} [w_{11}^{(1)} x_1 + w_{12}^{[1]} x_2]$$

$$f(w, x) = \frac{1}{1 + e^{-(w_0 x_0 + w_1 x_1 + w_2)}}$$

Computational graph.



$$w_0 x_0 = -2$$

$$w_1 x_1 = 6$$

$$w_2 = -3$$

$$\frac{1}{1 + e^{-(1)}}$$

$$\therefore e^{-1} = 0.3678$$

$$\approx 0.37$$

$$= \frac{1}{1.37}$$

$$= 0.729$$

$$\approx 0.73$$

differentiation of $1/x = (x^{-1}) = -x^{-2} = -\frac{1}{x^2}$

Local Gradient = 1.37

Global Gradient = 1

final Gradient $= -\frac{1}{(1.37)^2} * 1 =$

$= -\frac{1}{1.8769} = -0.53$

constant of this

$f = c + x$

$\frac{df}{dx} = 1$

final Gradient = local x Global

Local = 1

Global = -0.53

$= 1 \times 0.53(-1)$

$= -0.53$

$f = e^x$

$\frac{df}{dx} = e^x$

$\because x = -1 = e^{-1}$

$= 0.37$

final gradient = local x global.

$= 0.37 \times -0.53$

$= -0.1961$

≈ -0.2

$f = c - x$

$f = -1$

final = local x global.

$= -1 \times -0.2$

$= 0.2$

28/07/2026